

NXP CUP 2025

Tunisia's Journey to International Finals

AERONAR

Tunisia - First International Qualification

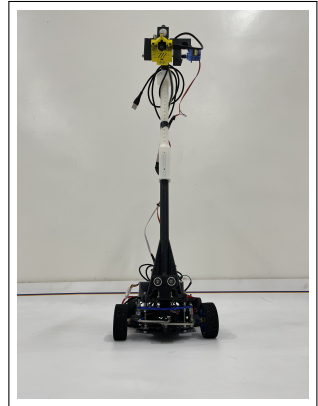
December 11, 2025

What is NXP Cup?

- **Autonomous lane-following robot competition**
- Based on camera + microcontroller
- Professional, international event
- Great learning opportunity for career development

Tunisia's Participation

Second year Tunisia joins
First Tunisian team qualified for international finals!



- **Not extremely difficult**, but teaches valuable skills
- Professional-grade project experience
- Opens doors to tech industry internationally
- NXP investing in Tunisia - bringing more teams

Let's make it count!

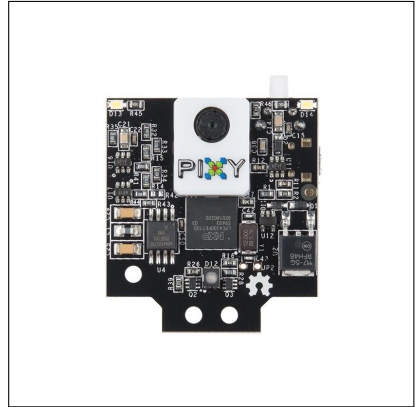
Pixy2 Camera Overview

What is Pixy2?

- Smart vision sensor
- Lane-detection camera
- Real-time object tracking
- Easy integration with microcontrollers

Key Features

Fast processing, configurable detection, robust performance



PixyMon Interface:

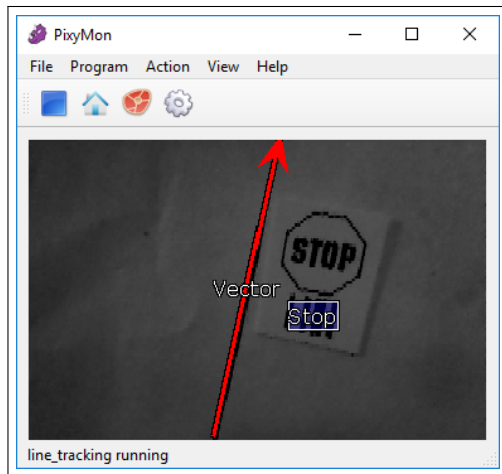
- Configuration tool for Pixy2
- Real-time camera view
- Signature tuning
- Parameter adjustment
- Live feedback during development



Critical Tool

PixyMon is essential for calibration and debugging!

Camera Output - Line Detection



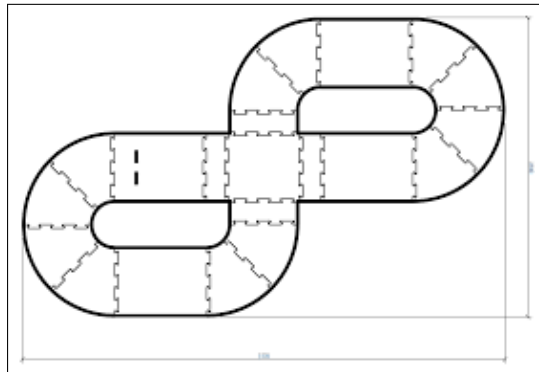
What You See

Track Challenges

LEGO Track

Both qualification and finals use LEGO tracks
Robot must stay in lane - errors accumulate quickly

- Precision is critical
- Small deviations compound
- Testing on similar tracks essential



- **600 RPM motors:** Good for Tunisia, adequate performance
- International teams often use higher speeds
- Balance between speed and control

Time Management

Start testing EARLY - like any competition, early testing = better results

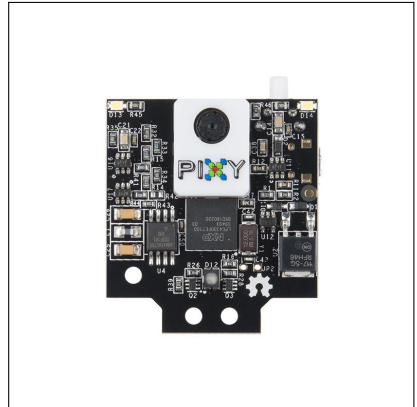
Pixy2 Tuning & Calibration

Tuning Experience:

- Good camera, but needs time to master
- Tuning is critical for performance
- Environment sensitivity matters
- Recalibration needed based on lighting
- Use PixyMon for real-time adjustments

Key Insight

Settings in PixyMon interface are crucial - spend time tuning!



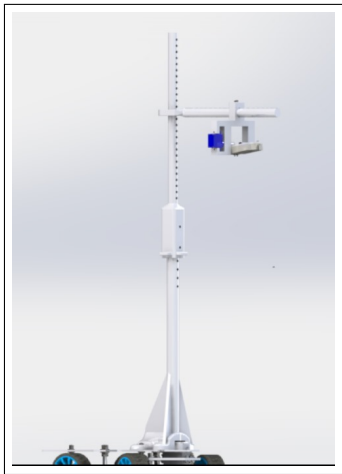
Don't Use Default Design Blindly

Adapt the design according to YOUR robot needs

- International teams ALL customize their designs
- **No one uses the Gitbook version as-is**
- Make it fit your components and strategy
- Iterate based on testing



Mechanical Design Examples



Communication is Essential!

Advanced Control & Debugging Features:

- **Remote control** - Manual override during testing
- **Remote debugging** - Monitor robot status in real-time
- **Single Board Computer (SBC)** - Raspberry Pi for advanced processing
- **Bluetooth module** - Wireless communication & parameter tuning
- **Serial logging** - Debug output for analysis

Why It Matters

Remote debugging saves hours of testing time and enables live tuning!

What International Teams Use

Advanced architectures from top teams:

- **Single Board Computers (Raspberry Pi) + NXP chip**
- Full **computer vision** pipelines
- **Live visualization** of camera view + robot reaction
- Amazing debugging tool during development
- Radio override (e.g., FS-IA6 module)
- Remote start/stop buttons

Key Insight

Even without SBC, plan your robot with debugging outputs & control override modules

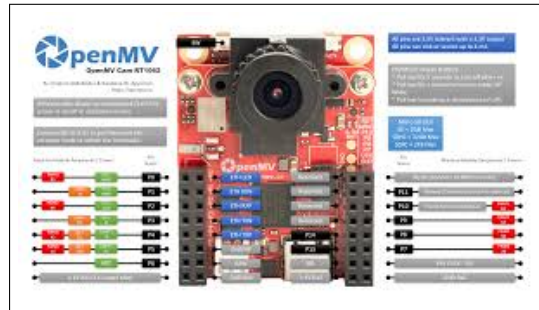
Camera Options

Pixy2 Camera:

- What we used
- Easy to use
- Good for beginners
- Vector-based line detection

OpenMV RT1062:

- Commonly used by international teams
- More powerful processor
- Full computer vision capabilities
- Python programmable



OpenMV RT1062 Camera

Helpful Links

Resource	Link
Pixy2 Documentation	docs.pixycam.com
Teensy 4.x Overview	pjrc.com/teensy
NXP Cup Official	nxp.com/nxp-cup
NXP Cup Gitbook	nxp.gitbook.io

These resources were essential for our journey

This is more than a university project

- Real chance to build pro-grade robot
- Opens doors to international tech industry
- Valuable career experience
- Represents Tunisia on global stage

Our Advice

Start early, debug well, and enjoy the ride!

#TeamTunisia — NXP CUP 2025

Questions?

Thank you for your attention!

Let's bring more Tunisian teams to NXP Cup!